**Nagaon Education Society's**
**GANGAMAI COLLEGE OF ENGINEERING, NAGAON,**
**DIST.-DHULE-05**
NAAC B+ Accredited

**KAVIYATRI BAHINABAI CHAUDHARI,**
**NORTH MAHARASHTRA UNIVERSITY, JALGAON**

**NES's**

**GANGAMAI COLLEGE OF ENGINEERING**
ISO 9001:2008

# Laboratory Manual

**Class:** T.E. **Semester:** _____

**Subject:** SE & AT

**Academic Year:** 2026-27.

**Subject Teacher:** Prof.Pradip P Dhole

## Institute Vision

- Empowering first generation engineers to excel technical education based on human values.

## Institute Mission

- To impart affordable and quality education in order to meet needs of industry and to achieve excellence in teaching learning process.
- To achieve excellence in application-oriented research in selected area of Technology to contribute to the development of the region.
- To collaborate with industries to promote innovation capabilities of budding engineers.
- To develop responsible citizens to awareness and acceptance of ethical values.
- To build a support system of all stakeholders to develop the institute.

# CERTIFICATE

This is to certify that Mr./Miss _____ of **Third year** student **Branch**_____, **Roll. No.**_____ has performed practical work satisfactorily in the subject **Practical based on SE & AT** in the premises of the Engineering College during year 2026–2027.

Date: / / Signature of the Teacher.

Place: Nagaon. Head of the Department.

# Table of Contents



| Sr. No | Experiment Title | Sign |
|---|---|---|
| 1 | Case study on SDLC models | |
| 2 | Preparation of Software Requirement Specification (SRS) | |
| 3 | UML diagrams for a given system. | |
| 4 | Introduction to Selenium tool | |
| 5 | Automation of test cases using Selenium | |
| 6 | Testing of web application using automation framework | |

**NES's GANGAMAI COLLEGE OF ENGINEERING, NAGAON**

Department of Computer Engineering

Name: _____ Date of Performance: _____

Class: TE Date of Completion: _____

Subject: SE & AT Roll No: _____

Sign of Subject Teacher: _____

## EXPERIMENT NO. 01

### Aim

To study different Software Development Life Cycle (SDLC) models and understand their phases, advantages, disadvantages, and applications through a case study

### Software Requirements

- Computer or laptop
- Internet connection
- MS Word or notebook for documentation

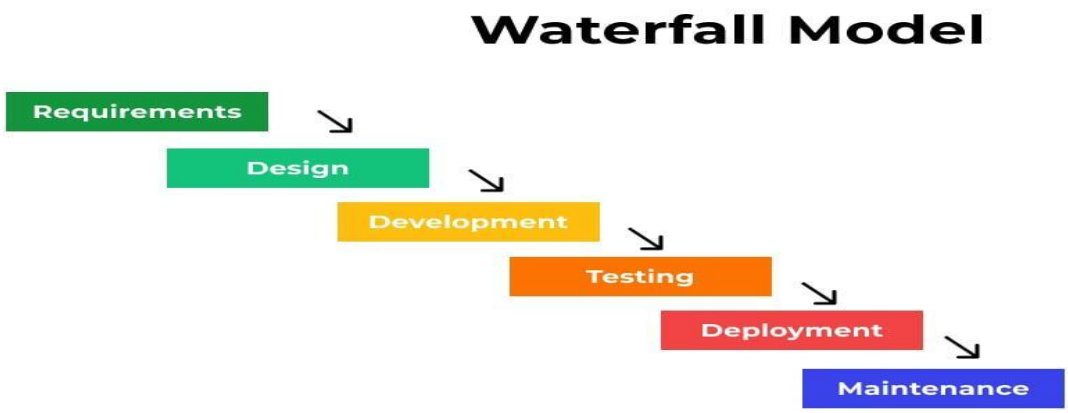
### Theory

SDLC (Software Development Life Cycle) is a systematic process used to develop high-quality software. It defines the different stages involved in planning, designing, developing, testing, deploying, and maintaining software.

The main phases of SDLC are:

1. Planning
2. Requirement Analysis
3. System Design
4. Implementation (Coding)
5. Testing
6. Deployment
7. Maintenance

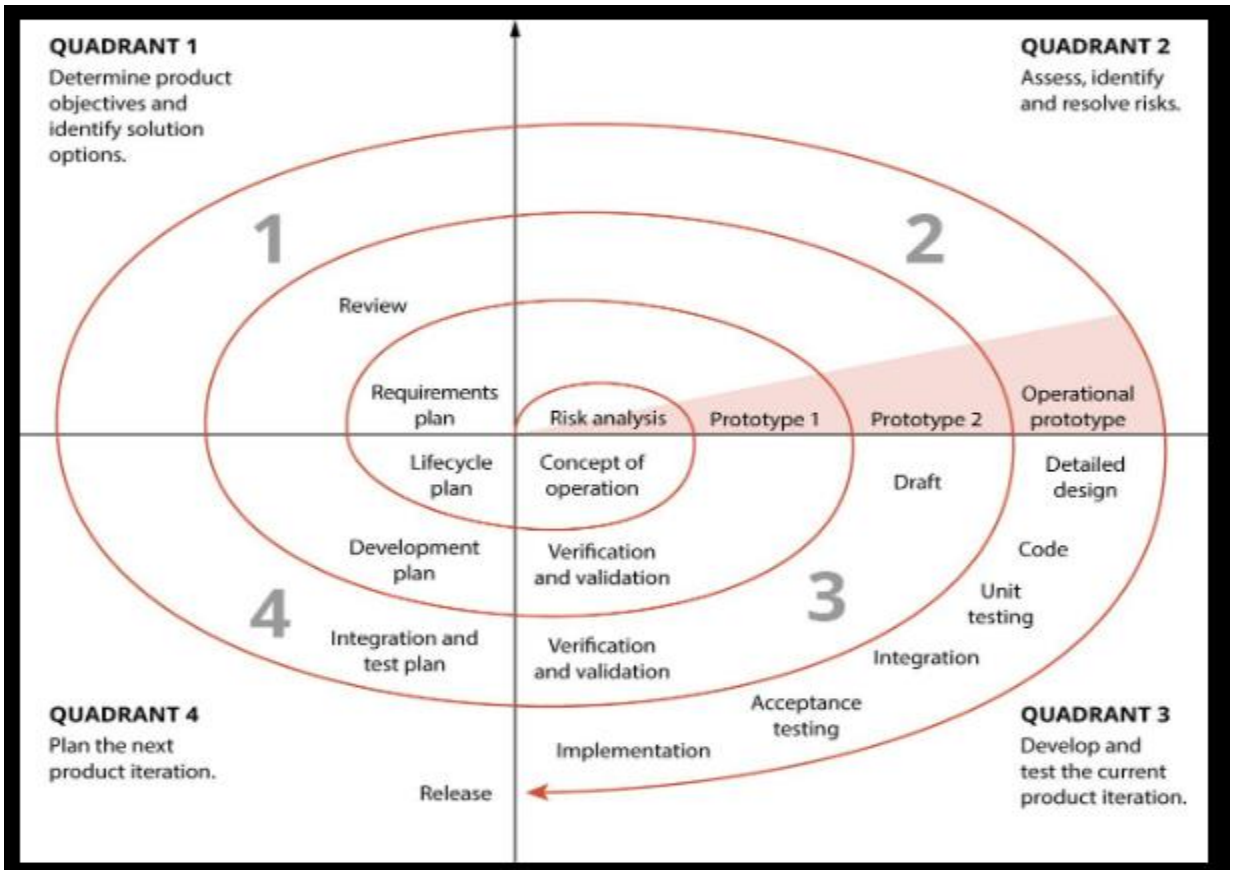
### Different SDLC Models




## 1. Waterfall Model

Development proceeds sequentially, with each phase generally completed before the next begins.

Example: A small college payroll system with clearly defined requirements



**2. Spiral Model**

Development occurs in cycles, with special attention to risk analysis.

Example: Banking software involving complex security requirements.

## Case Study: College Management System

### Problem statement

A college wants to develop a College Management System to manage student records, attendance, examination marks, and fee details.

### Objectives

- Store student information digitally.
- Maintain attendance records.
- Manage examination marks.
- Generate student reports.
- Reduce paperwork and manual errors.

### Applying the Waterfall Model



**Explanation of phases**

1. Planning: Decide the budget, development time, and project objectives.
2. Requirement Analysis: Collect requirements from teachers, students, and college administrators.
3. System Design: Design the database, user interface, and system architecture.
4. Coding: Develop modules for student registration, attendance, marks, and fees.
5. Testing: Check whether all modules work correctly and identify errors.
6. Deployment: Install and make the system available to college users.
7. Maintenance: Fix errors and update the software when required.

## Advantages of SDLC

1. Provides a structured development process.
2. Helps estimate project time and cost.
3. Improves software quality through testing.
4. Makes project planning easier.
5. Helps identify problems early.

## Disadvantages of SDLC

1. Some models require extensive documentation.
2. Changes can be difficult in strictly sequential models.
3. Testing and development may require significant time.
4. The wrong model can increase project cost and risk.

## Result

Different SDLC models were studied, and the Waterfall Model was applied to a College Management System case study to understand the stages of software development

**Questions:**

1. What is SDLC?
2. What is the main purpose of SDLC?
3. Name any four SDLC models.
4. Which model is suitable for changing requirements?
5. What is the testing phase?
6. What is software maintenance?

**NES's GANGAMAI COLLEGE OF ENGINEERING, NAGAON**

Department of Computer Engineering

Name: _____ Date of Performance: _____

Class: TE Date of Completion: _____

Subject: SE & AT Roll No: _____

Sign of Subject Teacher: _____

## EXPERIMENT NO. 02

### Aim

To prepare a Software Requirements Specification (SRS) document for a College Management System.

### Software Requirements

- Computer or laptop
- MS Word or any text editor
- Internet browser (optional)

### Theory

SRS (Software Requirements Specification) is a document that describes the functional and non-functional requirements of a software system. It explains what the system should do and the conditions under which it should operate.

An SRS document helps developers, testers, clients, and users understand the requirements before software development begins.

### Problem Statement

A college wants to develop a College Management System to manage student information, attendance, examination marks, and fee records. The system should reduce paperwork and make information easy to access.

### Objectives

1. Maintain student records digitally.
2. Manage attendance and examination marks.
3. Maintain fee payment records.
4. Generate student reports.
5. Provide secure access to authorized users

## Scope of the System

The College Management System will provide the following features:

- Student registration and profile management.
- Teacher and administrator login.
- Attendance management.
- Examination and marks management.
- Fee record management.
- Report generation.

## Functional Requirements

Functional requirements describe the specific functions that the system must perform.

| Requirement ID | Requirement | Description |
|---|---|---|
| FR01 | User Login | The system shall allow authorized users to log in. |
| FR02 | Student Registration | The system shall store student details. |
| FR03 | Attendance Management | Teachers shall record student attendance. |
| FR04 | Marks Management | Authorized staff shall enter examination marks. |
| FR05 | Fee Management | The system shall maintain fee payment records. |
| FR06 | Report Generation | The system shall generate student reports. |
| FR07 | Student Search | Users shall search for student records |

## Non-Functional Requirements

Non-functional requirements describe the quality and operating conditions of the system.

| Requirement | Description |
|---|---|
| Performance | The system should respond quickly to user requests. |
| Security | Only authorized users should access protected information. |
| Reliability | The system should operate consistently. |
| Usability | The interface should be easy to understand. |
| Maintainability | The software should be easy to update and maintain. |
| Backup | Student records should be backed up regularly. |

## Hardware Requirements

- Processor: Dual-core or higher
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB of available space
- Input devices: Keyboard and mouse

- Output device: Monitor

## Software Requirements

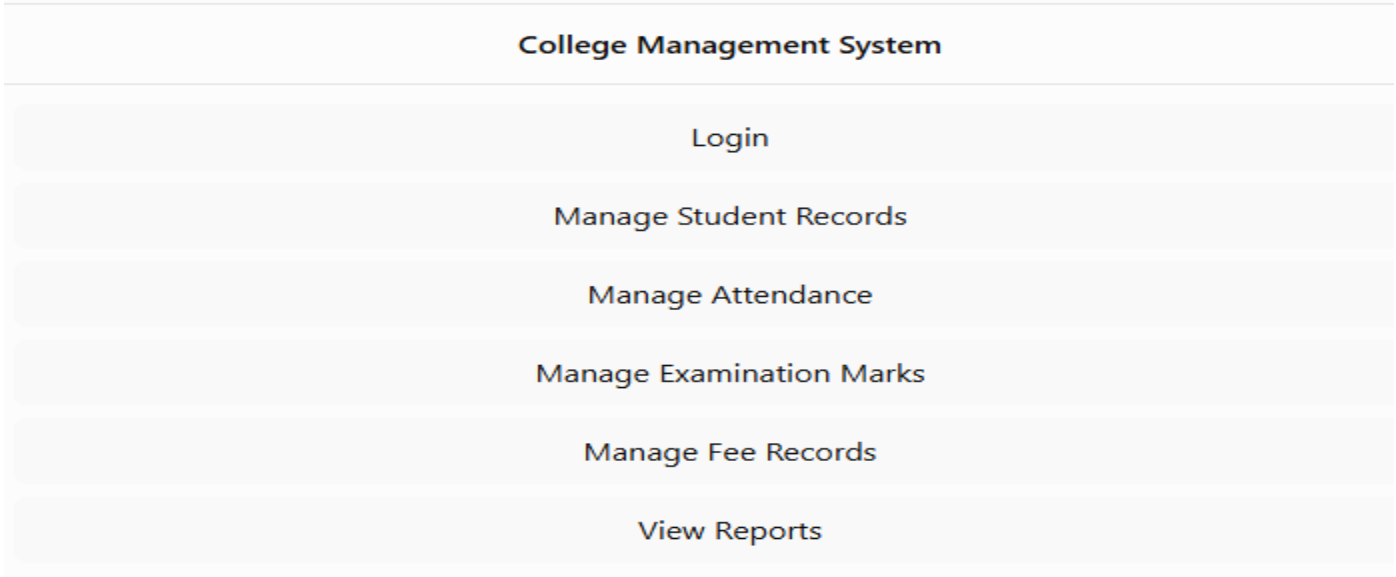
- Operating System: Windows or Linux
- Frontend: HTML, CSS and JavaScript
- Backend: Python or Java
- Database: MySQL
- Browser: Google Chrome or Mozilla Firefox

## System Users

The system has three main types of users:

1. Administrator: Manages users and college records.
2. Teacher: Records attendance and examination marks.
3. Student: Views personal details, attendance, and marks.

Use Case Description



## Constraints

1. Users must have valid login credentials to access protected features.
2. The system requires a suitable database.
3. Access to records must depend on user permissions.
4. Regular backups are necessary to protect information.

## Assumptions

- The college provides accurate student information.
- Teachers and administrators receive login credentials.
- Users have access to a computer and the required network.
- Authorized staff maintain the records.

## Expected Outcome

The system will maintain student records, manage attendance and marks, track fees, and generate reports efficiently.

## Result

The Software Requirements Specification (SRS) document for the College Management System was prepared, including its objectives, scope, functional requirements, non-functional requirements, users, constraints, and assumptions.

## Questions

Q1. What is SRS?

Q2. What is the purpose of an SRS document?

Q3. What are functional requirements?

Q4. What are non-functional requirements?

Q5. Who prepares an SRS document?

Q6. Why is SRS important in software testing?

**NES's GANGAMAI COLLEGE OF ENGINEERING, NAGAON**

Department of Computer Engineering

Name: _____ Date of Performance: _____

Class: TE Date of Completion: _____

Subject: SE & AT Roll No: _____

Sign of Subject Teacher: _____

## EXPERIMENT NO. 03

### Aim

To study and design UML diagrams, including Class Diagram, Component Diagram, and Deployment Diagram, for a College Management System.

### Software Requirements

- Computer or laptop
- Internet connection
- UML diagram tool: Draw.io (diagrams.net) or StarUML
- Web browser

### Theory

UML (Unified Modeling Language) is a standard visual modeling language used to represent the structure and behavior of a software system.

UML diagrams help developers understand the system before coding begins.

The three diagrams covered in this practical are:

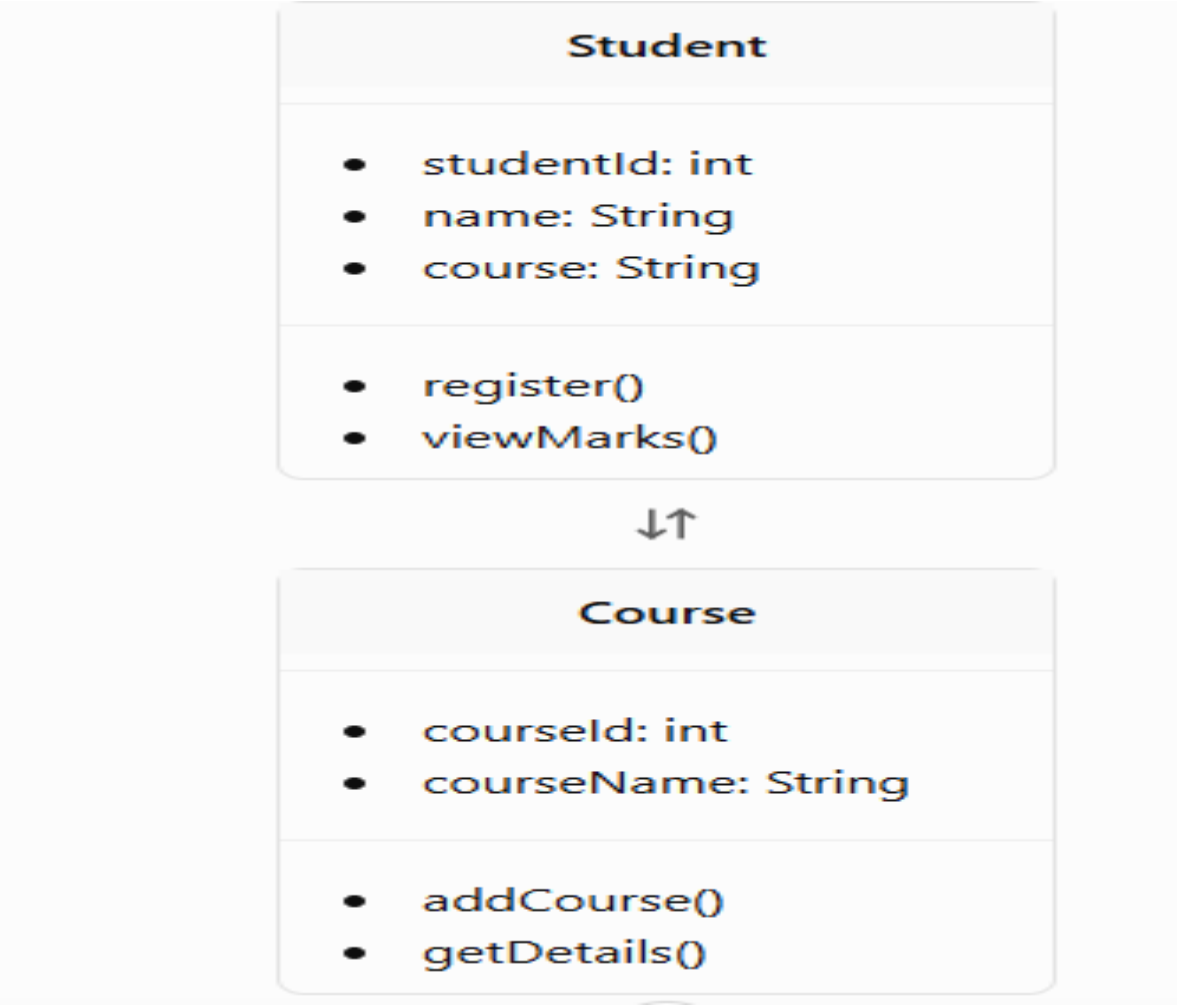
1. Class Diagram
2. Component Diagram
3. Deployment Diagram

### Case Study: College Management System

A college wants to develop a system to manage students, teachers, courses, and examination marks. UML diagrams are used to represent its software structure and deployment.

### Class Diagram

A Class Diagram represents classes, their attributes, methods, and relationships.

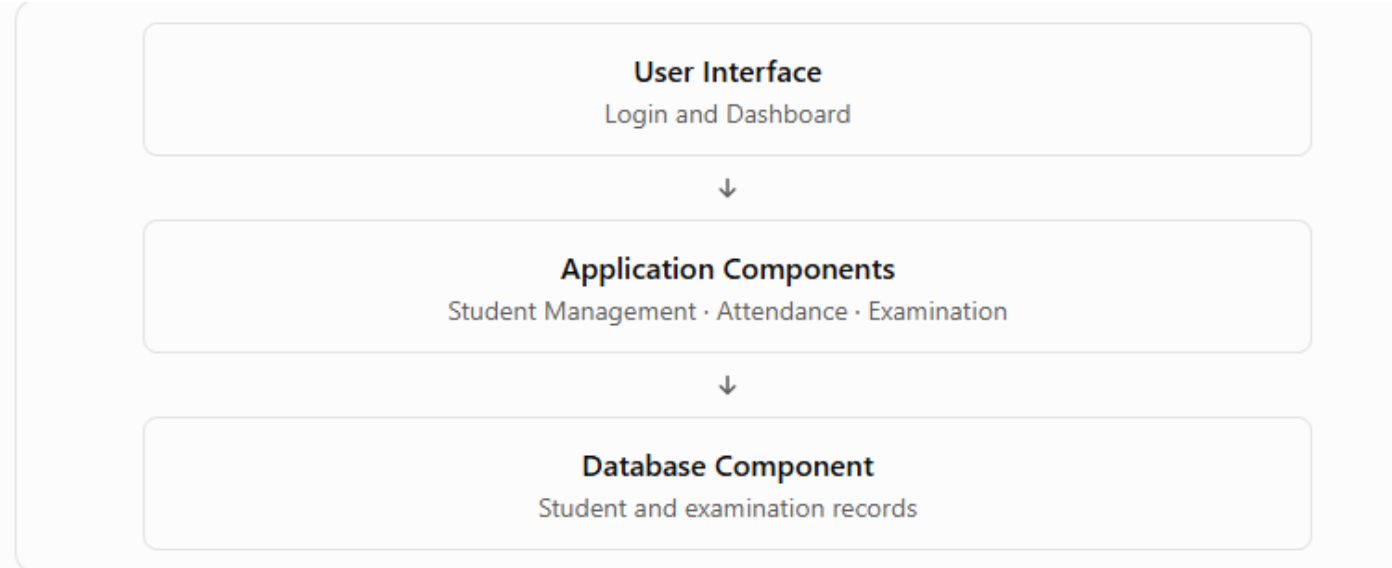



**Explanation**

- Class: Defines the structure of an object.
- Attributes: Store information about an object.
- Methods: Define the operations an object can perform.
- Relationship: Represents how classes are connected

## Component Diagram

A Component Diagram represents the major software modules and their dependencies.

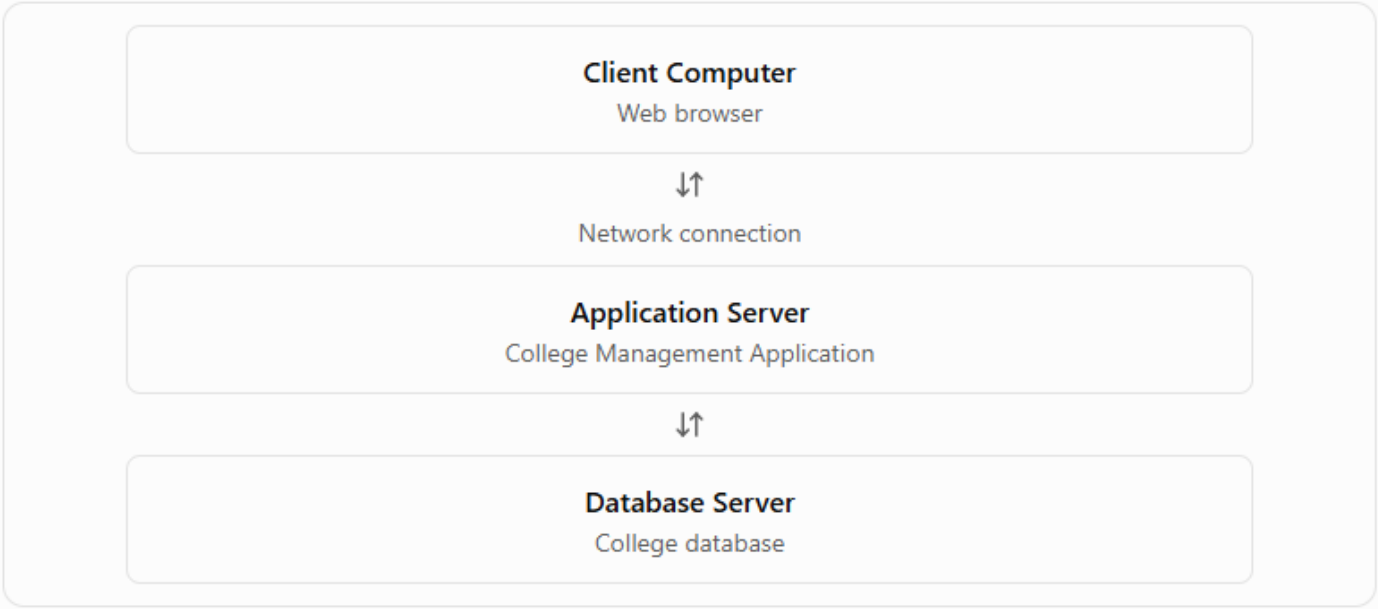



### Explanation

- User Interface accepts user input.
- Student Management manages student records.
- Attendance manages attendance information.
- Examination manages marks.
- Database stores and retrieves information

## Deployment Diagram

A Deployment Diagram shows the physical hardware nodes and the software deployed on them.




**Explanation**

- Client Computer: Used by students, teachers, and administrators.
- Application Server: Processes requests and runs application logic.
- Database Server: Stores college information.
- Network: Connects the client, application server, and database server.

## Procedure to Create UML Diagrams Using Draw.io

1. Open Draw.io.
2. Select Create New Diagram.
3. Choose a blank diagram.
4. Use rectangles to create classes and components.
5. Add names, attributes, and methods.
6. Draw connecting lines to represent relationships or dependencies.
7. Create a separate diagram for deployment nodes.
8. Save or export each diagram as a PNG or PDF file.

## Applications of UML

1. Software system design.
2. Understanding system architecture.
3. Communication between developers and clients.
4. Documentation of software projects.
5. Planning software before implementation.

## Questions

Q1. What is UML?

Q2. What is a Class Diagram?

Q3. What is a Component Diagram?

Q4. What is a Deployment Diagram?

Q5. Name one UML diagram tool.

Q6. What is an attribute?

Q7. What is a method?

**NES's GANGAMAI COLLEGE OF ENGINEERING, NAGAON**

Department of Computer Engineering

Name: _____ Date of Performance: _____

Class: TE Date of Completion: _____

Subject: SE & AT Roll No: _____

Sign of Subject Teacher: _____

## EXPERIMENT NO. 04

### Aim

To study the Selenium tool and use Python Selenium WebDriver to open a web browser, access a website, print its title, and close the browser automatically.

.

### Software requirements

- Operating System: Windows
- Python installed on the computer
- Visual Studio Code (VS Code)
- Google Chrome browser
- Selenium Python library

### Theory

Selenium is an open-source tool used to automate web browsers. It helps testers perform tasks such as opening websites, entering data, clicking buttons, and testing web applications automatically. Selenium WebDriver allows Python programs to control a browser

### Installation procedure

Step 1: Open VS Code.

Step 2: Open the terminal using Terminal → New Terminal.

Step 3: Install Selenium using this command

```
pip install selenium
```

Step 4: Create a Python file named `selenium_intro.py`.

Step 5: Write the following program

## Program

```
from selenium import webdriver

driver = webdriver.Chrome()

driver.get("https://www.example.com")

print(driver.title)

driver.quit()
```

## Explanation of the program

| Code | Explanation |
|---|---|
| `from selenium import webdriver` | Imports Selenium WebDriver. |
| `driver = webdriver.Chrome()` | Opens a Chrome browser session. |
| `driver.get("https://www.example.com")` | Opens the Example website. |
| `print(driver.title)` | Prints the title of the website in the terminal. |
| `driver.quit()` | Closes the browser session. |

## Execution procedure

1. Save the file as `selenium_intro.py`.
2. Connect to the internet.
3. Open the VS Code terminal.
4. Execute the following command:

   ```
   python selenium_intro.py
   ```

5. Observe Chrome opening the website.
6. Check the terminal for the website title.
7. Observe that Chrome closes after the program finishes.

## Expected output

The terminal should display:

```
Example Domain
```

## Questions

Q1. What is Selenium?

Q2. What is WebDriver?

Q3. What is the use of `driver.get()`?

Q4. What is the use of `driver.title`?

Q5. What is the use of `driver.quit()`?

**NES's GANGAMAI COLLEGE OF ENGINEERING, NAGAON**

Department of Computer Engineering

Name: _____ Date of Performance: _____

Class: TE Date of Completion: _____

Subject: SE & AT Roll No: _____

Sign of Subject Teacher: _____

## EXPERIMENT NO. 05

### Aim

To automate a test case using Python and Selenium WebDriver to open a website and verify whether the webpage title matches the expected title

### Software Requirements

1. Operating System: Windows
2. Python
3. Visual Studio Code (VS Code)
4. Google Chrome browser
5. Selenium library

### Theory

Selenium is an open-source tool used for automating web browsers. Selenium WebDriver allows us to control a browser using Python code.

Test case automation means executing test steps automatically instead of performing them manually.

In this practical, we will:

- Open a website.
- Retrieve its page title.
- Compare the actual title with the expected title.
- Display the test result.
- Close the browser.

### Installation

Open the VS Code terminal and execute:

```
pip install selenium
```

## Program

Create a Python file named `practical6.py`.

```
from selenium import webdriver

driver = webdriver.Chrome()

driver.get("https://www.example.com")

actual_title = driver.title

expected_title = "Example Domain"

if actual_title == expected_title:
    print("Test Case Passed")
else:
    print("Test Case Failed")

driver.quit()
```

## Explanation of the Program

| Statement | Explanation |
|---|---|
| `from selenium import webdriver` | Imports Selenium WebDriver. |
| `driver = webdriver.Chrome()` | Starts a Chrome browser session. |
| `driver.get(...)` | Opens the specified website. |
| `actual_title = driver.title` | Retrieves the actual webpage title. |
| `expected_title = "Example Domain"` | Stores the expected webpage title. |
| `if actual_title == expected_title:` | Compares the actual and expected titles. |
| `print("Test Case Passed")` | Displays success when both titles match. |
| `print("Test Case Failed")` | Displays failure when they do not match. |
| `driver.quit()` | Closes the browser session. |

## Execution Procedure

1. Open VS Code.
2. Create a file named `practical6.py`.
3. Write the above Python program.
4. Save the file.
5. Open the terminal.
6. Run the command:

```
python practical6.py
```

7. Observe the browser opening the website.
8. Check the test result in the terminal.

## Expected Output

```
Test Case Passed
```

The test passes when the actual webpage title matches `"Example Domain"`.

## Test Case Design

| Test Case ID | Test Scenario | Expected Result | Status |
|---|---|---|---|
| TC01 | Open the Example website | Website opens successfully | Pass |
| TC02 | Verify the webpage title | Title equals `Example Domain` | Pass |
| TC03 | Close the browser | Browser session terminates | Pass |

Note: These are the expected results when the website loads successfully and its title is correct.

## Advantages of Selenium Automation

1. Reduces repetitive manual testing.
2. Saves time when tests are repeated.
3. Improves consistency of test execution.
4. Helps identify failures automatically.
5. Supports browser automation using Python.

## Questions

Q1. What is test automation?

Q2. What is Selenium WebDriver?

Q3. What is an actual result?

Q4. What is an expected result?

Q5. What is an assertion?

**NES's GANGAMAI COLLEGE OF ENGINEERING, NAGAON**

Department of Computer Engineering

Name: _____ Date of Performance: _____

Class: TE Date of Completion: _____

Subject: SE & AT Roll No: _____

Sign of Subject Teacher: _____

## EXPERIMENT NO. 06

### Aim

To test a web application using Python, Selenium WebDriver, and the Pytest automation framework.

### Software Requirements

1. Operating System: Windows
2. Python
3. Visual Studio Code (VS Code)
4. Google Chrome browser
5. Selenium WebDriver
6. Pytest framework

### Theory

Selenium is a tool used to automate web browsers. Pytest is a Python testing framework used to write and execute test cases.

In this practical, we use Selenium with Pytest to open a website, verify its title, and automatically report whether the test passes or fails.

### Installation

Open the VS Code terminal and execute these commands:

```
pip install selenium
pip install pytest
```

## Program

Create a Python file named `test_website.py`.

```
from selenium import webdriver

def test_website_title():
    # Open Chrome browser
    driver = webdriver.Chrome()

    try:
        # Open the website
        driver.get("https://www.example.com")

        # Get the website title
        actual_title = driver.title

        # Verify the title
        assert actual_title == "Example Domain"

    finally:
        # Close the browser
        driver.quit()
```

## Explanation of the Program

| Code | Explanation |
|---|---|
| `from selenium import webdriver` | Imports Selenium WebDriver. |
| `def test_website_title():` | Defines a test function that Pytest can discover. |
| `webdriver.Chrome()` | Starts a Chrome browser session. |
| `driver.get(...)` | Opens the website. |
| `driver.title` | Retrieves the webpage title. |
| `assert actual_title == "Example Domain"` | Checks whether the actual title matches the expected title. |
| `finally:` | Ensures the cleanup code runs even if the test fails. |
| `driver.quit()` | Closes the browser session. |

## Execution Procedure

1. Open VS Code.
2. Create and save `test_website.py`.
3. Open the terminal in the same folder.
4. Run the test using:

```
python -m pytest -v test_website.py
```

5. Wait for Chrome to open the website and close.
6. Check the test result in the terminal.

## Expected Output

The terminal should display output similar to:

```
test_website.py::test_website_title PASSED
1 passed
```

## Test Case Design

| Test Case ID | Test Scenario | Expected Result |
|---|---|---|
| TC01 | Launch Chrome browser | Browser opens successfully |
| TC02 | Open the website | Example website loads |
| TC03 | Verify the webpage title | Title matches `Example Domain` |
| TC04 | Close the browser | Browser session terminates |

## Advantages

1. Tests can be executed automatically.
2. Repeated testing requires less manual effort.
3. Pytest provides clear pass/fail reports.
4. Selenium supports browser-based web application testing.
5. The framework can be extended with multiple test cases.

## Questions

Q1. What is an automation framework?

Q2. What is Pytest?

Q3. What is Selenium?

Q4. What is the purpose of the `assert` statement?

Q5. Why do we use `finally`?

Q6. How do we execute a Pytest program?